Advanced Control Systems

Assignment 1: DH-Table, Direct Kinematics, Inverse Kinematics, Geometric and Analytical Jacobian

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Abstract

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1 DH-Table

The Denavit-Hartenberg (D-H) parameter table is a standardized mathematical convention used in robotics to model the forward kinematics of serial-link manipulators. It systematically describes the relative spatial transformation between consecutive coordinate frames attached to each link along a kinematic chain using four scalar parameters: link length (a_i), link twist (α_i), link offset (d_i), and joint angle (θ_i).

The D-H convention is widely adopted because it significantly simplifies kinematic modeling and computation. By enforcing specific geometric rules for frame placement—such as aligning the x_i-axis with the common normal between adjacent joint axes—the general six degrees of freedom required for 3D spatial transformations are reduced to just four parameters per joint. This structured approach allows for the systematic construction of individual homogeneous transformation matrices $\mathbf{A}_i$:

$$\mathbf{A}_i = \text{Rot}_{z,\theta_i} \cdot \text{Trans}_{z,d_i} \cdot \text{Trans}_{x,a_i} \cdot \text{Rot}_{x,\alpha_i} \tag{1}$$

$$\mathbf{A}_i = \underbrace{\begin{bmatrix} \cos\theta_i & -\sin\theta_i & 0 & 0 \\ \sin\theta_i & \cos\theta_i & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\text{Rot}_{z,\theta_i}} \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\text{Trans}_{z,d_i}} \underbrace{\begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\text{Trans}_{x,a_i}} \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\alpha_i & -\sin\alpha_i & 0 \\ 0 & \sin\alpha_i & \cos\alpha_i & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\text{Rot}_{x,\alpha_i}}$$

At the end of this multiplications the final DH-Matrix will exactly like this:

$$\mathbf{A}_i = \begin{bmatrix} \cos\theta_i & -\sin\theta_i\cos\alpha_i & \sin\theta_i\sin\alpha_i & a_i\cos\theta_i \\ \sin\theta_i & \cos\theta_i\cos\alpha_i & -\cos\theta_i\sin\alpha_i & a_i\sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2}$$

Concatenating these matrices yields the overall transformation matrix $\mathbf{T}_0^n$, defining the exact pose of the end-effector relative to the base coordinate system:

$$\mathbf{T}_n^0 = \mathbf{A}_1(q_1)\mathbf{A}_2(q_2)\cdots\mathbf{A}_n(q_n) = \prod_{i=1}^{n} \mathbf{A}_i(q_i) \tag{3}$$

1.1 Extended Kinematic Chain: Base and Tool Transformations

Standard Denavit-Hartenberg parameters only compute the transformation from Frame 0 (F_0) to Frame n (F_n). In real applications, we need to relate the actual tool tip to the global workspace:

- **Base Frame (F_B):** The global reference system (e.g., workbench or room origin).
- **End-Effector Frame (F_E):** The frame attached to the Tool Center Point (TCP).

To compute the complete position of the TCP relative to the world, we multiply the joint transformation matrix $\mathbf{T}_n^0(q)$ by two constant matrices:

$$\mathbf{T}_E^B(q) = \mathbf{T}_0^B \cdot \mathbf{T}_n^0(q) \cdot \mathbf{T}_E^n = \mathbf{T}_0^B \cdot \left(\prod_{i=1}^{n} \mathbf{A}_i(q_i) \right) \cdot \mathbf{T}_E^n \tag{4}$$

1.1.1 Base Frame Transformation ($\mathbf{T}_0^B$)

Frame 0 is fixed on Joint 1 and does not always sit at the world origin F_B. The matrix $\mathbf{T}_0^B$ offsets the robot base by position (x_b, y_b, h_b) and rotation γ_b:

$$\mathbf{T}_0^B = \begin{bmatrix} \cos\gamma_b & -\sin\gamma_b & 0 & x_b \\ \sin\gamma_b & \cos\gamma_b & 0 & y_b \\ 0 & 0 & 1 & h_b \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{5}$$

Why it is used:

- (x_b, y_b, h_b) is the physical 3D distance from the world origin to the robot base.
- γ_b is the rotation angle of the robot mounting base.
- If the robot is moved to a new table, only $\mathbf{T}_0^B$ changes; internal robot parameters stay the same.

1.1.2 Tool Transformation ($\mathbf{T}_E^n$)

Frame n lies at the wrist flange, but tasks happen at the tool tip (TCP). The matrix $\mathbf{T}_E^n$ describes an extension L_t, a lateral shift d_x, and a tool rotation of $-90°$:

$$\mathbf{T}_E^n = \begin{bmatrix} 0 & 0 & -1 & d_x \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & L_t \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6}$$

Why it is used:

- $(d_x, 0, L_t)$ is the physical distance from the robot wrist to the end-effector tip.
- The rotation aligns the tool orientation (e.g., pointing the fingers forward).
- When replacing the tool (e.g., gripper to torch), only $\mathbf{T}_E^n$ needs to be updated.